MILITARY SPECIFICATION

RECEIVING SET, RADIO AN/TRQ-23( )

1. SCOPE

1.1 This specification covers the following units of one type of tactical radio communication intercept and direction finder designated Receiving Set, Radio AN/TRQ-23( ).

No. Required	Nomenclature
1	Antenna Group OE-4( )/GR
3	Bag, Sand
1	Case, Spare Parts
1	Clock, Aircraft Mechanical, 8 Day
2	Cord Assembly DC-307-A
1	Extinguisher, Fire
2	Garment Hanger
1	Gyro Magnetic Compass System J-2 Amplifier Electronic Control, Type A-2 Compensator, Magnetic Flux CN-405/ASN Control, Elec, Driven Gyro, Type S-3 Indicator, Induction Compass, Type V-8 Transmitter, Induction Compass T-611/ASN
1	Hammer, 4 lb.
2	Headset, Electrical H-113/U
1	Hose, Flexible
1	Installation Kit, Electronic Equipment MK-522/G Interconnecting Cables
1	Inverter Assembly, Rotary (J-2 Compass)
1	Lantern
1	Mount, Resiliant MT-3670( )/GR (for R-901( )/GR) Mount, Resiliant MT-3671( )/GR (for R-902( )/GR) Mount, Resiliant MT-( )/GR (for PP-4034( )/GR) Power Cable Assemblies

FSC 5895

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No. Required	Nomenclature
1	Power Distribution Panel
1	Power Supply PP-4034( )/TRQ-23
1	Rack, Special Purpose
2	Receiver Group OA-1451( )/PRR (Modified)
2	Receiver, Radio R-901( )/GR (p/o R-1418( )/GR)
1	Receiver, Radio R-902( )/GR (p/o R-1419( )/GR)
1	Receptacle, Ash
2	Recorder Reproducer AN/PNH-4
1	Rod, Ground
1	Seat Assembly, Operator's
1	Shelter, Electrical Equipment S-144( )/G
1	Signal Distribution Radio Panel SB-2250( )/TRD-22
3	Stake GP-2
2	Switch, Foot SA-825/TNH-5
2	Typewriters
2	Whip Antenna (Shelter Mounted)
1	Whip Antenna (Portable) AT-904/GR

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-516	Enamel, Lustreless, Quick-Drying Styrenated Alkyd Type
UU-T-111	Tape, Paper, Gummed (Sealing and Securing)
DDD-T-86	Tape, Textile, Cotton, General Purpose, (Unbleached, Bleached or Dyed)
GGG-H-86	Hammer, Hand (Forged Steel Head)

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PPP-B-20	Bags, Cotton, Mailing
PPP-B-35	Bag, Textile, Shipping, Burlap, Cotton and Waterproof Laminated
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)

MILITARY

MIL-E-75	Electron Tubes, Packaging, Packing, And Container Marking For (Package Group 1)
MIL-P-116	Preservation, Methods Of
MIL-B-117	Bags and Sleeves, Interior Packaging
MIL-C-3432	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-C-3767	Connectors, Electrical (Power, Bladed Type), General Specification For
MIL-C-3774	Crates, Wood, Open, 12,000-And-16,000 Pound Capacity
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-P-8585	Primar Coating, Zinc Chromate, Low-Moisture -Sensitivity

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MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-R-11461	Rod, Ground (With Attachments)
MIL-C-12520	Connector, Plug and Receptacle, (Electrical Waterproof), and Accessories, General Specification For
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-E-46061	Enamel, Comouflage, Solar-And Heat-Reflecting
MIL-S-52060	Shelter, Electrical Equipment S-144( )/G
MIL-R-55139	Receiver Group OA-1451( )/PRR
MIL-E-55301	Electromagnetic Compatability
MIL-R-55467(EL)	Receiver, Radio R-1418( )/GR
MIL-A-55468(EL)	Antenna Group OE-4( )/GR
MIL-R-55487(EL)	Receiver, Radio R-1419( )/GR

STANDARDS

FEDERAL

FED-STD-595	Colors

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MIL-STD-105	Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MS 75027	Cable, Cord, Standard Lengths and Standard Tolerances For Communication, Power and Radio Frequency Cable and Cord Assemblies.

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
SC-B-61956	Stake GP-2
SC-D-82768	Air Craft Loading Plate
SC-B-83314	Fastner, Latch
SC-DL-157329	Cable Assembly RF CG-409( )/U
SM-D-352594	Patch Cord Assembly (Red)
SC-C-539428	Bracket Assembly, Lantern
SC-C-539491	Lantern
SC-B-539528	Outlet, Convenience

ORDNANCE CORPS (Redstone Arsenal)

PART DRAWING

No. A-8192241	Clock, Aircraft Mechanical

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3. REQUIREMENTS

3.1 Description.- Receiving Set, Radio AN/TRQ-23( ) is a shelter mounted communication intercept and direction finding system capable of operation in the frequency range of 0.536 to 405 megacycles. It provides an intercept capability for both horizontally and vertically polarized signals from 0.536 to 405 megacycles and provides a direction finding capability on vertically polarized AM, CW, or MCW signals from 0.536 to 20.5 megacycles and AM, CW, MCW or FM signals from 20.0 to 405 megacycles. The set also has a direction finding capability for both vertically and horizontally polarized signals from 95 to 405 megacycles. Individual receiver and antenna assemblies may be removed from the shelter for man-transportable operation.

3.2 First Article (Preproduction Samples).- The contractor shall furnish preproduction samples for evaluation and approval as required by the invitation for bids and contract.

3.2.1 Preproduction test procedure.- Prior to start of preproduction testing the contractor shall furnish, to the Government, for approval, a complete electrical preproduction test procedure (4 copies). This procedure shall define detail tests for all requirements (see 3.25). The procedure shall include detailed test circuits and essential data on the test equipment, including manufacturer's model, serial number, date of latest calibration, and accuracy. When special test equipment is used, the test procedure will include schematic diagrams and a description of its operation and functions. Sufficient test data to show compliance with the above electrical requirements shall accompany the test procedure. The Government will require 30 days for evaluation of the test data and approval of the test procedure.

3.2.2 Reference standards.- After preproduction samples have been approved, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

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3.3 Procurement model.- A model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Except as otherwise specified herein or in the invitation for bids, the equipment shall conform to the following:

3.3.1 Physical construction.- Physical construction of the equipment shall conform to the model.

3.3.2 Features.- The equipment shall incorporate all features of the model.

3.3.3 Performance.- The contractor shall adjust and aline the model for optimum performance and then shall measure its performance characteristics. Performance of the equipment shall conform to these measurements, except that, in the event of conflict between specified performance characteristics and the performance of the model, specified performance characteristics shall govern.

3.4 Government furnished property.- When provided for in the bid request and contract, the following items shall be furnished by the Government:

Shelter, Electrical Equipment S-144/G
Receiver Group OA-1451A/PRR
Gyro, Magnetic Compass System J-2 (Components of)
Inverter Assembly, Rotary (For J-2)
Recorder Reproducer AN/PNH-4
Typewriter
Heat and Light Power Kit MK-522/G
Foot Control SA-825/TNH-5
Rack, Special Purpose MT-2345/G
Mast Section MS-116
Mast Section MS-117
Mast Section MS-118
Mast Base AB-15/GR
Antenna AT-904/GR
Fire Extinguisher
Headset, Electrical H-113/U
Cord Assembly CD-307-A

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3.5 Parts, materials and processes; general.- In addition to the requirements of this specification, the requirements of Specification MIL-P-11268, including the selection requirements therein, shall apply.

3.5.1 Wiring.- All hook-up wiring used in chassis or shelter shall be considered as being in sealed inclosures containing heat dissipating parts (Reference Para 3.2.2.2 of MIL-P-11268).

3.5.2 Finish, protective.- Equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware and touch-up after mounting, where required. The final paint film on interior walls, ceilings, doors and duct work shall be light green semigloss enamel matching color chip 24533 of Federal Standard FED-STD-595. Floors shall be painted lustreless ocean gray color chip 36118 of Federal Standard FED-STD-595. The J-2 compass system, fire extinguisher, heater, receiver, inverter, recorder-reproducer, foot switch, and typewriter shall not be refinished. Except as otherwise specified, all other internally mounted equipment shall be semigloss light gray color chip No. 26250 of Federal Standard FED-STD-595. The final paint film on items which are stored within the shelter but which are normally used externally to the shelter such as the Flux Gate Guard and the Whip Antenna Assemblies, shall be olive drab semigloss enamel matching color chip X-24087. The final paint film on the following items shall also be olive drab semigloss enamel matching color chip X-24087: Receiver Mounts, Signal Distribution Panel and Power Supply (PP-4034).

3.5.3 Marking.-

3.5.3.1 General.- Marking shall conform to Specification MIL-M-13231. Front Panel marking shall be Group 1 as described in that specification.

3.5.3.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

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3.5.3.3 Serial numbers.- The following items shall be serial numbered:

Power Supply PP-4034( )/TRQ-23
Receiving Set Radio, AN/TRQ-23( )
Receiving Set R-744( ) (Modified)
Signal Distribution Radio Panel SB-2250( )/TRQ-23

3.5.3.4 Aircraft loading data plates.- The aircraft loading data plate for Government furnished Shelter S-144( )/G shall be removed and replaced with a plate depicting the loading data for Receiving Set, Radio AN/TRQ-23( ) in accordance with Specification MIL-M-13231 and Electronic Command Drawing SC-D-82768.

3.6 Shelter modification and installation.- Installation of all equipments into the shelter shall be made by the contractor. The installation includes all necessary modification to the shelter, mounting provisions (brackets, hardware, etc.) cabeling, lighting and power facilities. The performance characteristics of the installed equipment shall not be impaired by the installation. All external modifications to the shelter shall be weatherproof and RF protected. After completion of modification the shelter shall still be capable of meeting requirements of Para 3.17 of MIL-S-52060.

3.6.1 Shelter refinishing.- The outer surface of the shelter shall be refinished as follows:

c. After cleaning, prime with one (1) coat of zinc chromate primer, 0.4 to 0.6 mil (dry film) thick per Specification MIL-P-8585.

b. Apply one mil intermedicte film of white lustreless enamel conforming to Federal Specification TT-E-516.

c. Apply one mil final film of dark green lustreless enamel per Specification MIL-E-46061.

3.6.2 Labeling.- The following label, meeting the requirements of Specification MIL-M-13231, shall be affixed to the refinished surface:

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"WARNING - This surface has been painted with a special solar reflecting paint which has the effectiveness of a white surface. Do not repaint except as follows: For touch-up of final film, use dark green lustreless enamel per Specification MIL-E-46061, FSN 8010-905-7133. For touch-up of intermediate film use white lustreless enamel per TT-E-516, FSN 8010-878-5761."

3.7 Shelter, weight distribution and mobility.- With all components installed, and stowed for shipment, weight distribution shall be such that approximately an even load is placed on all corners of the shelter. Lateral unbalance, if necessary, will be permitted up to 10 percent of the gross weight of the assemblage (shelter and payload) with the road side the heavier. Fore and aft unbalance shall not exceed 5 percent of the payload of the shelter with the front of the shelter the heavier. All assemblages shall have adequate roadability and be capable of rapid movement over highways and cross country type terrain including steep slopes, rough surfaces, mud and snow. The maximum weight of the AN/TRQ-23 shall not exceed 1500 pounds.

3.8 Receiver Group OA-1451( )/PRR (Modified).- This equipment shall be furnished by the Government for modification and incorporation into the AN/TRQ-23( ) system. Receiver (R-744A/PRR) shall be modified to provide a diode output in accordance with the procurement model. After modification the receiver shall meet requirements of Para 3.8b, sheets (3, 5, 6, 7, 8, 9 and 10) of MIL-R-55139A at room temperature and after subjection to the temperature and moisture resistance tests in Para 4.10.1 and 4.10.2 to insure that the modification has not degraded receiver performance.

3.8.1 Diode output Receiver R-744( )/PRR.- A minimum of 2.5 volts diode output into an impedance of 50,000 ohms shall be produced by the signal level specified for minimum AM sensitivity as per Specification MIL-R-55139 for all service conditions. The polarity of the output shall be negative with respect to ground.

3.9 Receivers, Radio R-901( )/GR and R-902( )/GR.- These receivers shall conform to Specifications MIL-R-55467(EL) and MIL-R-55487(EL) respectively.

3.10 Rack, special purpose.- Two racks will be furnished by the Government. These racks shall be modified in accordance with the procurement model and utilized to mount the major portion of the electronic equipment in accordance with the procurement model.

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3.11 Gyro magnetic compass system J-2.- The components of this compass system will be furnished by the Government to be utilized in lieu of the compass in the procurement model. The indicator portion of the J-2 compass shall be mounted on Signal Distribution Panel SB-2250( )/TRQ-3 as shown in Figure 1. The remaining components, except the flux gate sensing element, shall be mounted to the rack behind Receiver R-901( )/GR and Antenna Control C-6183/GR. The flux gate sensing elements shall be installed in accordance with the procurement model. The flux gate mast shall be mounted on a pedestal which may be electrically disconnected and stowed above the operators' seats in the shelter, raised to its full height, electrically connected and keyed to orient itself to the heading of the vehicle or electrically disconnected in a lowered position. The accuracy, installation and finish of the flux gate mast shall be in accordance with the model, except that an appropriate non-magnetic deflecting guard shall be mounted on the shelter roof forward of the flux gate for the purpose of preventing damage to the flux gate against the hazards of low overhanging objects such as limbs and branches of trees while the flux gate is in the lowered position. The deflecting guard and the mast shall be easily removable for air transportation and stowed in the shelter. Interconnections and wiring for the J-2 system shall be furnished by the contractor. A cover plate for the flux gate opening in the shelter shall be provided by the contractor to retain the rainproof feature of the shelter. The cover plate shall be stowed on the shelter roof as per model when not in use except wing nuts shall be used instead of screws.

3.11.1 Inverter Assembly, Rotary.- This item will be furnished by the Government. Its purpose is to provide 400 cycle power for Compass J-2. The inverter shall be mounted in accordance with the model.

3.12 Recorder Reproducer AN/PNH-4( ).- Two (2) of these recorder reproducers will be provided by the Government. Installation of these items shall be in accordance with the procurement model.

3.12.1 Switch, Foot SA-825/TNH-5.- Switches shall be furnished by the Government for operation of Recorder Reproducer AN/PNH-4. For storage one switch shall be clamped on the cab end between the opening at the bottom of the rack, the other switch shall be clamped on the bracket to the left side of the power supply. When stowed, these switches shall not interfere with the storage of any other item.

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3.13 Installation Kit, Electronic Equipment MK-522/G.- This kit will be furnished by the Government for heating and lighting purposes. Applicable portions of this kit shall be utilized as specified in 3.2.4.1 and 3.2.4.2. Items in this kit which are not utilized in the AN/TRQ-23( ) shall be returned to the Government intact.

3.14 Signal Distribution Radio Panel SB-2250( )/TRQ-23.- This panel shall provide a means to interconnect the RF output of the shelter mounted or remote antenna subassemblies to the appropriate radio receiver, to connect the appropriate receiver diode output to the Indicator, Azimuth IP-740/TRQ-23, to connect the receiver audio output to the appropriate operating position and to connect the appropriate Antenna Control to the Antenna Drive. The signal distribution panel layout shall conform to Figure 1 in lieu of the layout in the model. All instruments in the panel shall be suitably shock mounted to withstand mechanical service conditions specified herein. RF Connectors which receive patch cables shall be color coded. RF receptacles shall be coded orange. Whip receptacles shall be coded yellow. RF patch cables shall be identified with a 1/2 inch band of colored tape at each end of the cable. Two cables shall be orange and two yellow. Four audio patch cables, 18 inches long per Drawing SM-D-352594, shall be provided. The diode patch cord shall be in accordance with the model except that it shall be colored green.

3.14.1 Clock, aircraft, mechanical.- An 8 day mechanical clock shall be furnished in accordance with Ordnance Part Drawing A-8192241. It shall be incorporated in the signal distribution panel as shown in Figure 1.

3.15 Power distribution panel.- The power distribution panel shall control the input power to the shelter, the 24 volt output of Power Supply PP-4034( )/TRQ-23 and distribution of the 24 volt output to the shelter circuits. The 110 volt, a.c. outlet shall be in accordance with SC-B-539528.

3.15.1 Input voltage.- The input voltage to the power distribution panel shall be from a 115/230 volt 47 to 63 cycle a.c. power source and a nominal 27 volt vehicular power source. A toggle lock switch shall allow selection of either power source. The a.c. power source shall be used to supply Power Supply PP-4034( )/TRQ-23 whose output shall provide a nominal 27 volts to the power distribution panel.

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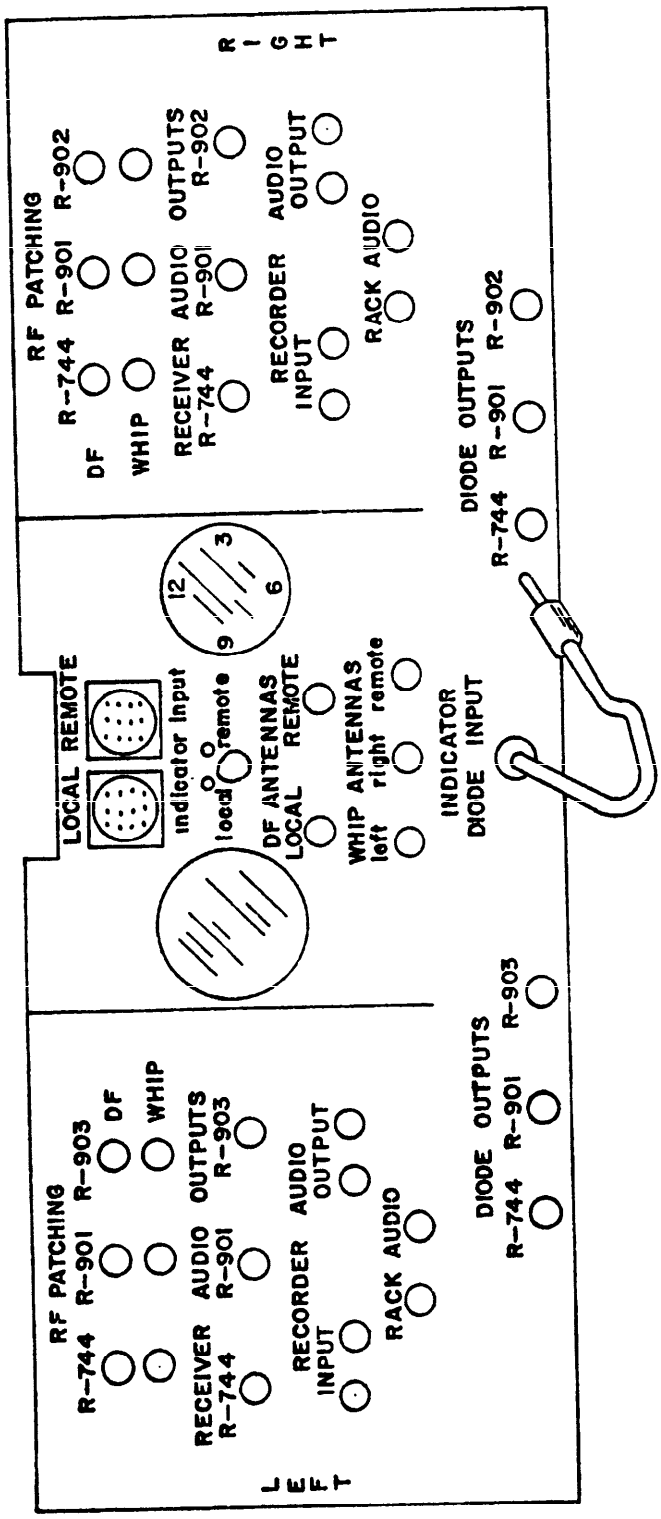


FIG. 1 SIGNAL DISTRIBUTION PANEL, RADIO SB-2250 ( ) /TRQ-23

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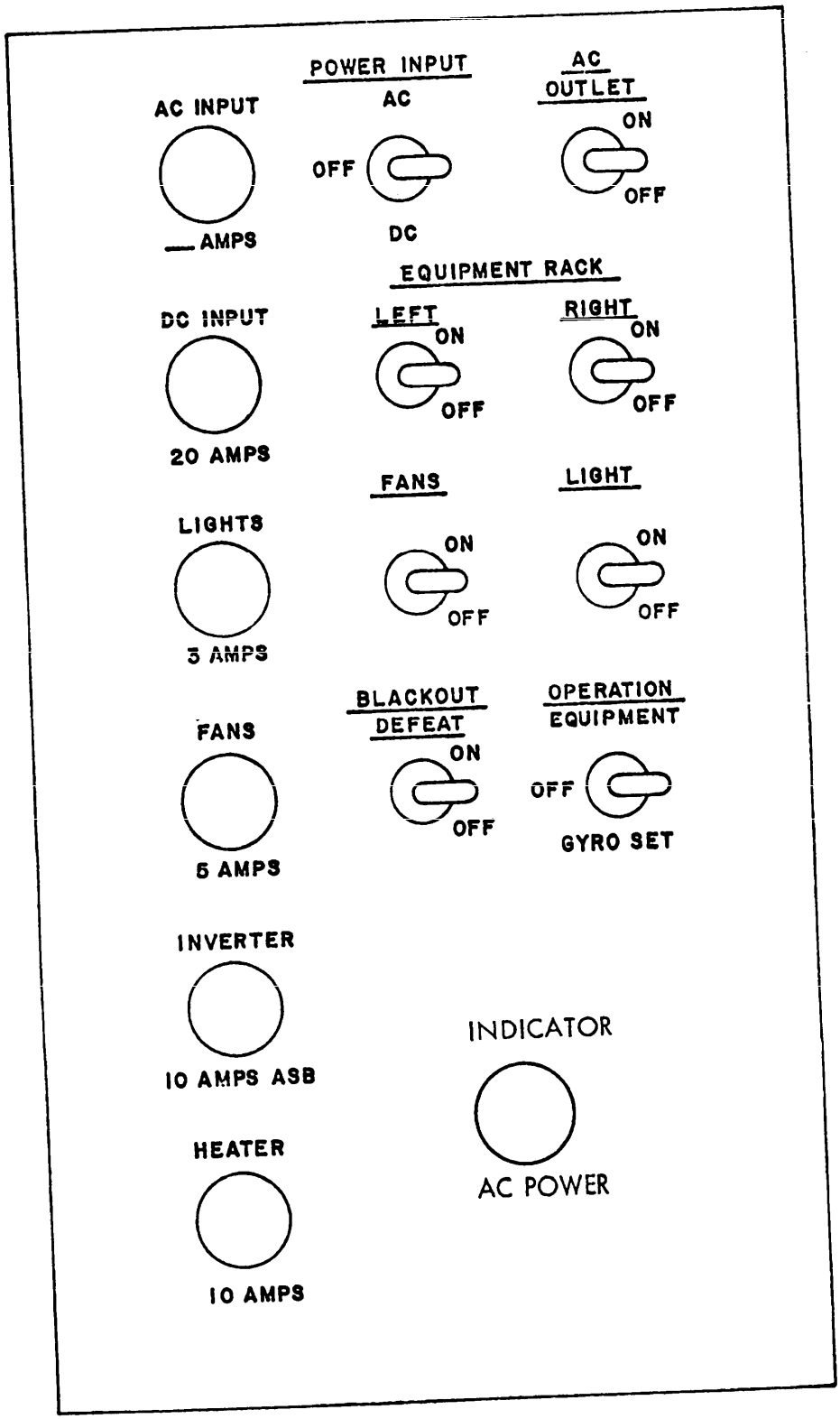


FIG. 2 POWER DISTRIBUTION PANEL

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3.15.2 Panel switching controls.- All switches on the panel shall be located and their positions marked as shown in Figure 2. The shelter circuits shall be controlled by the following switches to be located on the power distribution panel:

a. Power input.- A toggle lock on-off-on switch shall allow selection of the a.c. voltage to the Power Supply PP-4034( )/TRQ-23 (and its output to the shelter) or the 24 volt vehicular power source.

b. AC outlet switch.- An on-off switch to allow energizing of the 110 volt a.c. power receptacle located on the power distribution panel.

c. Right rack switch.- An on-off switch to control the power to the following equipments located on the right side of the rack: Receiver R-744( )/PRR, R-901( )/GR, R-902( )/GR, Recorder Reproducer AN/PNH-4 and Control Unit C-6183( )/GR.

d. Left rack switch.- An on-off switch to control the power to the following equipments located on the left side of the rack: Receiver R-744( )/PRR, R-901( )/GR, Control Unit C-6183( )/GR, Recorder Reproducer AN/PNH-4 and Azimuth Indicator IP-740/GR.

e. Fan switch.- An on-off switch to provide power to the shelter ventilating fans.

f. Light switch.- An on-off switch to provide power to the shelter lights.

g. Blackout defeat switch.- An on-off switch to allow the lights to remain on when the shelter door is opened.

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h. Equipment-Gyro set switch.- An on-off-on switch to allow selection of shelter equipment, excluding the gyro set or selection of the gyro set and lights only.

i. Electrical protection.- The power distribution panel shall contain fuses for protection of the circuits. All input voltages to the shelter shall be fused. Each 24 volt electrical circuit in the shelter shall be protected with fuses in accordance with the model. The d.c. input shall be protected against reversed voltage. A red light shall be provided on the power panel to indicate that a.c. power is available.

3.16 Mechanical safety.- The equipment shall be such that when assembled and set up for operation, accidental contact by personnel with moving parts such as gears, fans and belts shall be prevented. Projecting and overhanging edges shall be held to a minimum Edges and corners of enclosures shall be rounded.

3.17 Antenna Group OE-4( )/GR.- Antenna Group OE-4( )/GR shall conform to the requirements of Specification MIL-A-55468(EL).

3.17.1 Fixed pedestal.- The fixed pedestal shall be mounted in the shelter roof in accordance with the procurement model. Reinforcement of the shelter roof is required to support the pedestal and to adequately support the operator's weight. A deflecting guard shall be provided forward of the pedestal to protect the drive shaft and connector from damage due to low overhanging branches. The fixed pedestal and deflecting guard shall be removable for airlift capability and stowed in the shelter. A cover plate shall be provided by the contractor to cover the pedestal opening when the pedestal is not installed. The cover plate shall preserve the rainproof characteristics of the shelter and shall be stowed in the shelter when not in use in accordance with the model. The fixed pedestal housing shall be stowed in the shelter in accordance with the model when not in use.

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3.17.2 Whip antennas.-

3.17.2.1 Shelter mounted whip antennas.- The following items will be furnished by the Government to be utilized in the fabrication and installation of the shelter mounted whip antennas:

2 each Mast Sections MS-116( )
2 each Mast Sections MS-117( )
2 each Mast Sections MS-118( )
2 each Mast Base AB-15/GR

The contractor shall fabricate a mast bracket for each shelter mounted whip antenna in accordance with the model. The mast bracket shall allow the antenna to be controlled from a vertical to a horizontal position from the operator's position inside the shelter. The control handle shall have a mechanical pin lock in the horizontal and vertical positions to lock the antenna in place. Installation of the antennas and the hardware shall be in accordance with the model. Two (2) RF cables and terminations shall be furnished by the contractor in accordance with Drawing SC-DL-157329 and shall be 1 foot 6 inches long. The mast base assemblies and antennas shall be removable for storage within the shelter. The RF cables shall be stowed in the spare parts case. The shelter mounted RF receptacles, whip orientation mechanism and whip clamp, shall be redesigned to provide a flush surface for the shelter exterior when the mast base assembly is removed. The RF input connector shall be provided with a cover and chain.

3.17.2.2 Stowage case, whip antenna.- A canvas stowage case shall be supplied for stowage of the antenna whips in the shelter. The case shall be in accordance with the model and shall conform to MIL-P-11268. They shall be stowed on the shelter wall above the operator's seat.

3.17.2.3 Portable whip antenna.- Antenna AT-904/GR will be furnished by the Government for use as a portable whip. This antenna, in its carrying cases shall be stored in the shelter when not in use.

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3.18 Power Supply PP-4034( )/TRQ-23.- The power supply shall convert the 50/60 cycle, 115/230 volt a.c. power to the nominal 27 volt d.c. necessary to operate the equipment in the shelter. An on-off-on lock type toggle switch shall be provided on the power supply to select either the 115 volt or 230 volt a.c. source. Fuse protection shall be provided for voltage inputs of 115 and 230. A shock mount with captivated wing nuts shall be provided for mounting the power supply in the shelter.

3.18.1 Input voltage requirements.- The power supply shall be unregulated and provide switched taps to increase or decrease the output voltage between 24 and 30 volts when operating over an input voltage range 100 to 130 volts or 210 to 250 volts a.c. single phase 47 to 63 cycles.

3.18.2 Output voltage requirements.- The output of the power supply shall be capable of being maintained between 24 and 30 volts while operating at input voltages specified in 3.18.1 and loads from 0 to 20 amps current. The output ripple shall not exceed 0.1 volts within its operating range.

3.18.3 Protective circuits.- The output of the power supply shall be protected from overheating and excessive current. The power supply shall turn off automatically when 32 to 35 volts output is exceeded. It shall be provided with a 20 amp circuit breaker capable of handling the starting current of the gyro compass system.

3.18.4 Power cable assemblies.- One (1) fifty (50) foot a.c. power cable and one (1) ten (10) foot d.c. power cable shall be supplied. The fifty (50) foot a.c. cable shall be for 50-60 cycles, 115/230 volt input; the ten (10) foot d.c. power cable shall be for 24 volts d.c. input. Cable length tolerances shall conform to Military Standard MS75027.

3.18.5 Connectors.-

a. The vehicle cable, ten foot d.c. power, shall be cord type CO-02MGF(2/12)SJ0480 per MIL-C-3432 with connector MS3108E22-2P and MS3057128 clamp per MIL-C-5015 on one end and connector UW1220FA17 per MIL-C-12520/5A on the other. The receptacle in the shelter for this power cable shall be UW2020MA00 per MIL-C-12520/8A with cover

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and chain per MIL-C-12520/12A, style 2X. Wire pin A for +24 V, and pin B for negative on both connectors. On MS3108E22-2P pin C is unused. On UW1220FA17 and UW2020MA00 provide jumpers between pins A and D and between B and C.

b. The fifty foot a.c. power cable shall be cord type CO-03HOF(3/18)0410 per MIL-C-3432 with connectors UP121M per MIL-C-3767 on one end and UW1220FA15 per MIL-C-12520/5A on the other. The shelter receptacle for this cable shall be type UW2020MA00 per MIL-C-12520/8A with cover and chain per MIL-C-12520/12A, style 2X. On connector UW1220FA13 the high voltage pair shall go to pins A and B; the shelter ground shall be wired to pin C.

c. The power supply input connector shall be type UW2020MA00 per MIL-C-12520/8A and the output connector shall be type UW2020FA00 per MIL-C-12520/9A.

3.19 Interconnecting cables.- The shelter cabling shall conform to cabling provided in the procurement model except that the 24 volt d.c. input power connector shall be mounted on the cab side of the shelter (see 3.18.5a). Appropriate clamps are required on all internal cabling.

3.20 Operator's seat.- The operator's seat shall be constructed in accordance with the model. Seat belts shall be provided for each operator and cushions shall have a sufficient number of air breathers on the exposed sides. The material used for the seat cushions shall be in accordance with the model.

3.21 Case, spare parts and accessories.- A suitable metal case for storing operating spare parts, two rolls of typewriter paper, CRT tube, hand tools and ancillary items, including batteries, shall be permanently installed under the operator's seat on the road side of the shelter. The case shall extend from the front wall (cab end) of the shelter, to one foot from the flux gate mast and shall not interfere with operation of the mast nor shall it interfere with the raising or lowering of the operator's seat. The case shall have two hinged covers. One cover shall extend from the cab end of the shelter to the door side of the first operator's

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seat and the other cover shall extend from there up to one foot of the flux gate mast. Each cover shall have two latches in accordance with Drawing SC-B-83314 for fastening down. The case shall contain special compartments as required and shall be sufficiently rugged to support a weight of 500 pounds. The case shall be painted lustreless gray color chip 36118 of Federal Standard FED-STD-595.

3.22 Typewriters.- The typewriters will be furnished by the Government for installation in accordance with the procurement model. Paper shall be fed from a roll located under the pull shelf to the rear of each typewriter.

3.23 Accessory items.-

3.23.1 Fire extinguisher.- A 5-pound CO_2 type fire extinguisher will be furnished by the Government and mounted in accordance with the procurement model.

3.23.2 Stake GP-2.- Three (3) each GP-2 stakes shall be supplied and stored in the shelter. These stakes are to be used to secure the portable antenna drive during severe wind conditions. They shall be stowed in the spare parts case under the operator's seat. The stakes shall be in accordance with SC-B-61956.

3.23.3 Bags, sand.- Three (3) burlap bags, 22 inches wide and 40 inches long, which when filled with sand or rock shall be used to hold down the portable antenna when stakes cannot be used. These bags shall be furnished in accordance with Federal Specification PPP-B-35, type I, grade 3, style A and stowed in the spare parts case.

3.23.4 Receptacle, ash.- An ash receptacle shall be supplied and permanently attached within the shelter. The receptacle shall be constructed of a metal material and a means shall be provided for readily emptying its contents without the use of tools. It shall be mounted in the center of the rack beneath the R-902( )/GR. The removable part shall remain secure during road tests.

3.23.5 Garment hanger.- Suitable garment hangers shall be provided and installed in an accessible location for hanging the two operator's outer garments, helmets, masks, weapons, canteens and pistol belts. Location of the hangers shall not interfere with operation of the equipment.

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3.23.6 Headset H-113/U.- The headset and Cord Assembly CD-307-A will be furnished by the Government and stored in the spare parts case.

3.23.7 Hammer, 4 pound.- A four (4) pound sledge shall be supplied for use in driving hold-down stakes. The sledge hammer shall be in accordance with Federal Specification GGG-H-86, type X, class 1. The sledge shall be firmly fastened in the spare parts case for stowage.

3.23.8 Tools.- The contractor shall provide tools necessary for initial set up and operation of the AN/TRQ-23.

3.24 Installation and stowage of items into the shelter.- Installation and stowage of items shall be in accordance with the model unless otherwise specified. Provision shall be made to secure all loose equipment to the shelter floor to insure against damage during specified service conditions. Securing hardware holding stowed items in the shelter shall be designed for loosening by hand.

3.24.1 Lighting provisions.- Lighting facilities shall be installed in the shelter in accordance with the procurement model except that both ceiling lights shall be located one foot closer to the curb side of the shelter. The contractor shall utilize applicable portions of Government furnished MK-522/G installation kit for this purpose. The lights shall be controlled by a blackout switch on the shelter door. A lantern and mounting bracket shall be provided in accordance with SC-C-539491 and SC-C-539428. It shall be installed in the shelter for fixed position use or removable for hand held use. The lantern shall be mounted on the shelter wall, curb side between the operator's seat and the fire extinguisher.

3.24.2 Heating and ventilating provisions.- Heating and ventilating facilities shall be installed in the shelter in accordance with the model. The contractor shall utilize the applicable portions of Installation Kit, MK-522/G for this purpose. When activated, the performance of the installed facilities shall be equal to that of the model. Items not used for installation in the shelter shall be returned to the Government. A five (5) foot length of flexible metal extension hose shall be supplied by the contractor for the heater exhaust. Metal guards shall be provided to prevent accidental actuation of the heater switches. A provision shall be made for stowing this hose inside the shelter during air transport. This hose shall be in accordance with the procurement model. Additional ventilating louvers

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shall be installed on the road side of the shelter wall. They shall be constructed of the same material and provide the same ventilating area as the louver located in the rear door of the procurement model. The additional louvers shall not affect fording capabilities of the shelter as defined in MIL-S-52060 or degrade equipment performance. Suitable filtering shall be installed in the heater circuits to prevent RFI interference. The acoustic noise generated by the fans, heater, and antenna drive when operating at maximum capacity shall not exceed the noise level of the procurement model.

3.24.3 Grounding.- A ground rod type II, style 3, five (5) feet long shall be provided by the contractor for grounding the shelter body. A terminal shall be provided on the shelter having a 3/8 inch stud with a captive wing nut for connecting to the ground strap. All components within the shelter shall be connected to the shelter ground. The ground rod shall meet requirements of MIL-R-11461 and shall be stored in the shelter.

3.25 System Operation.- The AN/TRQ-23 shall be energized continuously without failure for 56 hours at ambient temperature and normal a.c. line voltage. Performance of all electrical accessories, lights fans and heater shall be equal to the model. The gyro set heading error shall not exceed $\pm 5^{\circ}$ when energized. The system shall meet the acceptable performance levels established in accordance with Para. 3.3.3 for every intercept and DF combination shown in Figure 3.

3.26 720 hour test.-(Power Supply PP-4034( )/TRQ-23) When tested in accordance with 4.8; a failure occuring within the first 72 hours, or if more than 3 failures occur during the remainder of the test, this will be cause for rejection. A failure shall be defined as the need for maintenance other than replacement of lamps. The daily tests and tests at the end of the run shall meet specified requirements.

3.27 Preconditioning.- Power Supply PP-4034( )/TRQ-23 shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce and electrical preconditioning of Para. 4.9.

3.28 Service conditions.- The AN/TRQ-23 shall meet the following service conditions; where a test is referenced, meeting the test shall be considered as compliance with the requirements:

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Test	Antenna	Drive, Antenna	Control, Antenna C-6183 ( )/GR	Recorder Reproducer AN/PNH-4 -4	Receiver	Whip Antenna Shelter and Remote	Azimuth Indicator	Phone Jack
1	MX-6337( )/GR	MX-6335( )/GR	Right	1 Right	R-901 Right	Right	IP-740( )/GR	Right
2	MX-6338( )/GR	MX-6344( )/GR	Left	2 Left	R-901 Left	Left	IP-740( )/GR	Left
3	MX-6339( )/GR	MX-6335( )/GR	Right	1 Right	R-744 Right	AT-904/GR Remote	IP-740( )/GR	Right
4	MX-6340( )/GR	MX-6344( )/GR	Left	2 Left	R-744 Left	Left	IP-740( )/GR	Left
5	MX-6341( )/GR	MX-6335( )/GR	Left	1 Left	R-744 Left	Left	IP-740( )/GR	Left
6	MX-6342( )/GR	MX-6344( )/GR	Right	2 Right	R-902	Right	IP-740( )/GR	Right
7	MX-6343( )/GR	MX-6335( )/GR	Right	1 Right	R-902	Right	IP-740( )/GR	Right

FIGURE 3

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3.28.1 Temperature.- (See 4.10.1 and 4.7 for Electrical system test and requirements.)

(a) Operating: Ambient temperature in the range of +150° F to -40° F. (the 150° F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160° to -65° F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.28.2 Moisture resistance (see 4.10.2).- Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.28.3 Elevation (see 4.10.3).-

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

3.28.4 Orientation (see 4.10.16).-

(a) Operating: Any orientation up to 20 degrees from normal operating position (that is: forward, backward, left or right).

(b) Nonoperating: Storage in any position for a period of two years. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the preproduction sample shall be considered as compliance with this requirement.

3.28.5 Rainfall (see 4.10.4).- Two inches of rainfall per hour, with wind at 20 miles per hour.

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3.28.6 Sand and dust (see 4.10.5).- As encountered during heavy traffic over dry clay roads or in desert areas. (Applicable to Power Supply PP-4034( )/TRQ-23 only).

3.28.7 Vibration, bounce and shock.-

3.28.7.1 Vibration, internal (see 4.10.7).- With the equipment shock-mounts (if any) blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment, shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. Power Supply PP-4034( )/TRQ-23 only.

3.28.7.2 Vibration, equipment (see 4.10.8).- The natural frequency of the equipment, mounted on its shock-mounts, shall be between 25 and 35 cycles per second. The amplitude of vibration of the mounted equipment shall not exceed three times the amplitude of the applied vibration at any frequency between 10 and 25, or between 35 and 55 cycles per second. Power Supply PP-4034( )/TRQ-23 only.

3.28.7.3 Bounce and shock (see 4.10.9, 4.10.10, 4.10.11, and 4.10.11.1).- Units shall be capable of meeting the requirements of Table I and, for the purpose of selecting a bounce test, shall be classified as follows:

TABLE I - BOUNCE AND SHOCK

Paragraph Heading	Paragraph Number	Performance after Test
Bounce (Power Supply PP-4034( )/TRQ-23	4.10.9	Specified performance. No physical damage.
Munson Bounce (AN/TRQ-23)	4.10.10	Specified performance. No physical damage.

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TABLE I - BOUNCE AND SHOCK (Continued)

Paragraph Heading	Paragraph Number	Performance after Test
Road test, Perryman Course (or approved equal) AN/TRQ-23	4.10.14	Specified performance. No physical damage.
Shock; bench handling Power Supply PP-4034( )/TRQ-23	4.10.11.1	Specified performance. No physical damage.
Shock; railroad humping (AN/TRQ-23	4.10.11	Specified performance. No physical damage.

3.28.8 Interchangeability (see 4.10.15).- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when approved by the Contracting Officer may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.28.9 Electromagnetic compatibility (see 4.10.17).- The equipment, installed and operating in the shelter, shall meet the emanation and susceptibility limits of Specification MIL-E-55301 for tactical equipment. Application of suppression components and techniques shall conform to that specification.

3.28.10 Adjustment and repair (see 4.10.20).- The equipment shall be so constructed that parts, terminals, wiring, etc., are accessible for circuit checking, adjustment, maintenance, repair, and replacement with minimum disturbance to other parts and wiring and with use of the minimum number and variety of special tools, particularly those needed for tuning and adjustment. When tools for tuning or adjustment are needed, the unit where they are used shall provide convenient means for mounting them.

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3.28.11 Technical literature, tools and running spare parts.- Technical literature, tools, and running spare parts shall be furnished if specified in this contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.1, h, i).

3.28.12 Workmanship (see 4.10.20).- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

In Specification MIL-P-11268:

- General requirements for plastic material and plastic parts.
- Wiring and cabling, including:
  - Slack.
  - Protection.
  - Clearance.
  - Splicing.
  - Connections, general.
  - Shielding on wire and cable.
- Flux and cleaning agent for soldering.
- Process for soldering.
- Cleaning.
- Riveting.
- General requirements for securing of parts.
- Cleaning of welding.
- Process for welding.
- Mechanical operation of Controls.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery (see 4.3).

(b) Acceptance inspection:

(1) Acceptance inspection of equipment before preparation for delivery (see 4.4).

(2) Acceptance inspection of preparation for delivery (see 4.4.1).

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table III, the inspection specified for Group A, Group B and Group C (see Table II, III and IV, respectively). The preproduction inspection will normally be performed in this order: (1) Vibration, (2) Bounce, (3) Shock, Bench handling, and (4) Rain; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

TABLE II - PREPRODUCTION INSPECTION

Inspection (For additional preproduction inspection see 4.3) x	Requirement Paragraph	Inspection Paragraph
Orientation	3.28.4	4.10.16
Sand and dust	3.28.6	4.10.15
Internal vibration	3.28.7.1	4.10.7

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TABLE II - PREPRODUCTION INSPECTION (Continued)

Inspection (For additional preproduction inspection see 4.3)	Requirement Paragraph	Inspection Paragraph
Acoustic noise	3.24.2	4.10.16
Bounce (Munson Bounce)	3.28.7.3	4.10.10
Road Test (Perryman Course)	3.28.7.3	4.10.14
Railroad Humping (Shock Test)	3.28.7.3	4.10.11
Weight Distribution Test (Shelter)	3.7	4.10.12

4.4 Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Cable assemblies, audio and power	3.18.4
Finish	3.5.2
Brazing	3.5
Grounding to Chassis	3.24.3

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Item	Where required
Marking	3.5.3
Bags, sand	3.23.3

4.5 Acceptance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with any specification requirements. The Government will review and evaluate the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection to verifty the contractor's compliance with specified requirements (see 6.8). Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to Group A, Group B, and Group C inspection, except preproduction samples, shall be bounce preconditioned after final assembly. (See 3.27)

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table III and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE III - GROUP A INSPECTION

Inspection	Rqt. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical				
Cable Assemblies, Power	3.18.4	4.11	1.0%	4.0%
Power Supply, PP-4034( )/TRQ-23	3.18	4.10.11.1	1.0%	4.0%

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TABLE III - GROUP A INSPECTION (Continued)

Inspection	Rqt. Para.	Insp. Para.	AQL Major	AQL Minor
Case, Spare Parts	3.21	4.10.13	1.0%	4.0%
Ancillary Items	3.14.1, 3.17.4, 3.23.2, 3.23.7, 3.24.3, 3.17.3, 3.11, 3.23.8	4.11	1.0%	4.0%
Shelter, S-144( )/GR and installation of all items	3.10, 3.6, 3.6.1, 3.15.2, 3.24, 3.24.2, 3.18.3, 3.24.1, 3.7	4.11	1.0%	4.0%
Modification to Receiver, Radio R-744( )/PRR	3.8	4.6.3	1.0%	4.0%
Signal Distribution Panel	3.14	4.11	1.0%	4.0%
Power Distribution Panel	3.15	4.11	1.0%	4.0%
Electrical				
Power Supply PP-4034( )/TRQ-23	3.18	4.6.1	1.0%	*
Voltage Output	3.18.2	4.6.1.1		
Operating Voltages	3.18.1	4.6.1.2		

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TABLE III - GROUP A INSPECTION (Continued)

Inspection	Rqt. Para.	Insp. Para.	AQL Major	AQL Minor
Receiver R-744( )/PRR	3.8	4.6.3	1.0%	*
Diode Output	3.8.1	4.6.3		
System Operation	3.25	4.7	1.0%	*

*All Electrical Tests are Major

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table IV and to the special procedures for small-sample inspection of Standard MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met Group A inspection.

TABLE IV - GROUP B INSPECTION

Inspection	Rqt. Para.	Insp. Para.	AQL (see 4.5.2)
Interchangeability	3.28.2	4.10.15	4.0%

4.5.2.1 Group B sampling plans.- The group B sampling plans, for the AQL's listed in Table IV, shall be as follows:

AQL	Inspection level for normal, reduced and tightened inspection
4.0%	S-3

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4.5.3 Group C inspection.- This inspection shall be as listed in Table V, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection.

TABLE V - GROUP C INSPECTION

Inspection	Requirement Paragraph	Inspection Paragraph
720 Hour Test (PP-4034( )/TRQ-23( )	3.2	4.8
Temperature	3.2.8.1	4.10.1
Moisture Resistance	3.28.2	4.10.2
Elevation	3.28.3	4.10.3
Rain	3.28.5	4.10.4
Equipment Vibration	3.28.7.2	4.10.8
Bounce, Vehicular Mounted (PP-4034( )/TRQ-23)	3.28.7.3	4.10.9
Electromagnetic Compatability	3.28.9	4.10.17
Shock, Bench Handling (PP-4034( )/TRQ-23)	3.28.7.3	4.10.11.1

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4.5.3.1 Sampling for inspection of equipment.- Two items of production (excluding preproduction samples) for each Group C inspection shall be selected from the first 50 items produced and from each 100 items produced thereafter. The items shall be selected without regard to their quality and shall be selected from the first units produced of each group. Two units are required to permit different tests to be conducted at the same time. Both units are not to be subjected to the same tests.

4.5.3.2 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (I) the process and materials, as applicable, and, (II) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.6 Unit tests.-

4.6.1 Power Supply PP-4034( )/TRQ-23.-

4.6.1.1 Output voltage.- With the specified loads applied across the output terminals, the output voltage shall be measured with a 0.1% d.c. voltmeter for conformance to the requirements listed in 3.18. The output ripple shall be measured with a Hewlet Packard Model #120B or equal.

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4.6.1.2 Operating voltages.- During the test, the operating voltages shall be measured with a 0.1% voltmeter to assure compliance with the requirements listed in 3.18.

4.6.2 Shelter S-144( )/G.- The modified shelter shall be examined for compliance with the model requirements of 3.6, 3.6.1, 3.7, 3.24, 3.24.1 and 3.24.2.

4.6.3 Receiver R-744( ).- The modified receiver shall meet the requirements of Paragraphs 3.8 and 3.8.1.

4.7 Electrical system tests.-Unless otherwise specified the system electrical tests shall be conducted as follows:

a. All electrical accessories, lights, fans and heater shall be activated continuously.

b. The Gyro Set heading shall be checked for accuracy when energized. The gyro cannot be accurately checked in a chamber or building. Therefore this test will have to be made at a remote site.

c. The contractor shall establish acceptable performance levels by making electrical measurements, when the system has been adjusted for optimum performance, for every phase of the seven intercept and DF combinations shown in Figure 3 of this specification. Time required for this test shall not exceed 4 hours.

4.8 720 hour test.- Power Supply PP-4034( )/TRQ-23 shall be operated continuously at 27 volts, 20 amps for a period of 720 hours. At the end of each 24 hour period and at the end of the 720 hour run the unit shall be tested in accordance with Para. 4.6.1, and shall meet the requirements of Para. 3.18.1, 3.18.2 and 3.18.3.

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4.9 Preconditioning.-

4.9.1 Electrical burn-in time.- After final alignment and prior to preliminary electrical inspection each PP-4034( )/TRQ-23 on order shall be operated at 27 volts, 20 amps for 50 hours. The contractor shall make all necessary repairs and realignments to correct any failure during and at the completion of this burn-in time.

4.9.2 Bounce.- Power Supply PP-4034( )/TRQ-23, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.10 Service condition tests.-

4.10.1 Temperature test (see 3.28.1).- With doors and louvers open, the AN/TRQ-23 less the gyro, typewriters and recorder reproducer shall be subjected to steps 1, 2a, 3, 5, 7, 8 and 10 of the temperature cycle shown on Standard MIL-STD-169. Performance tests shall be conducted after steps 1, 3, 5, 6, 8 and 10 in accordance with Para. 4.7 for the AN/TRQ-23 and Para. 3.18 for the PP-4034( )/TRQ-23. Receivers R-901( )/GR and R-902( )/GR shall not exceed the 6 db degredation allowable under the operating vs temperature requirements of their respective specifications.

4.10.2 Moisture-resistance test (see 3.28.2).- The AN/TRQ-23 less the gyro, typewriters and recorder-reproducer shall be tested as follows:

4.10.2.1 Test conditions.-

(a) Do not remove equipment from humidity chamber for measurements.

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(b) Start measurements not more than 5 minutes after power is applied to equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Perform the entire test with shelter door and ventilating louvers open.

4.10.2.2 Test method.-

(a) Dry at 130° ±5° F for 24 hours.

(b) Condition at 77° ±5° F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.7, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.7 shall be made during the periods shown on the standard and shall comply with 3.25.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5° F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 3.25.

4.10.2.3 Performance.- Measurements made during test and specified performance during cycling shall be in accordance with full requirements of Para. 4.7 for the AN/TRQ-23 and Para. 3.18 for the PP-4034( )/TRQ-23.

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4.10.2.4 Failure.- If any equipment fails to meet the performance specified during cycling, it shall be realined or readjusted once. If the equipment then fails to meet (3.25) or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.10.2.2e, it does not pass the test.

4.10.3 Elevation test (see 3.28.3).-

(a) Elevation (10,000 feet). Receiving Set, Radio AN/TRQ-23 (less gyro , recorder reproducers and typewriters), shall be placed in a test chamber and the pressure shall be reduced to an absolute pressure of 20.58 inches of mercury. The equipment shall be operated at the reduced pressure for a period of at least 15 minutes with all controls set for maximum voltage. During this period the equipment shall give no indication of arcing, puncturing, flashover or breakdown and shall meet full specification requirements of Para. 4.7 and 3.18.

(b) Elevation (nonoperating) (50,000 feet). The equipment shall be placed in a test chamber and the pressure reduced to an absolute pressure of 5.54 inches of mercury. After exposure to this pressure for a period of at least 15 minutes, the equipment shall be removed from the test chamber and operated at maximum voltage to determine whether the reduced pressure has adversely affected the equipment so as to cause arcing, puncturing, flashover or breakdown.

4.10.4 Rain test (see 3.28.5).- The shelter, without the use of any external sealing, caulking, taping, etc., shall be subjected to the following two part test with doors and louvers closed:

(a) Each of the five exposed surfaces of the shelter, with doors and covers closed, shall be sprayed with water from nozzles, Model G295Q (or GG29SQ) as made by Spraying Systems Company, Bellwood, Illinois, or equal, pointing directly at the surfaces

FIGURE 7 Source: https://assist.dla.mil -- Downloaded: 202640-26T04:17Z
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and arranged to provide uniform coverage of the surface. Each end panel shall be tested using four nozzles, each side panel using six nozzles, and the roof using four nozzles. Each nozzle shall operate at the pressure of forty PSIG at the nozzle and shall be located twenty inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. Each panel shall be sprayed for a period of forty minutes. (If the rain test is to be followed by the Fording Test, then, the bottom row of nozzles can be eliminated with the exception of the nozzles facing the door-end panel.)

(b) Each end panel of the shelter, with louver covers open, shall be sprayed with water from four nozzles as used in (a) above. Two nozzles shall be located across the width of the end panel, thirty inches above the shelter and thirty-six out from the end panel and aimed at the nearest cover. The other two nozzles shall be located to each side of the end panel, thirty inches above the roof, twelve inches out from the end panel, and thirty-six out from the side panel and aimed at the louver open opening(s). Each nozzle shall be sprayed for a period of twenty minutes.

After the test, there shall be no evidence of leakage into the shelter or into the walls, floor, roof or doors. To determine that no water has entered the area between the skins, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed.

4.10.5 Sand and dust (see 4.6).- Power Supply PP-4034( )/TRQ-23 shall be tested as follows:

(a) Place the equipment in its normal operating position, in the test chamber. Raise and maintain the sand and dust density at 0.1 to 0.5 gram per cubic foot within the test space. The relative humidity shall not exceed 30 percent at any time during the test.

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(b) The sand and dust shall be of angular structure and shall have a particle size such that the particles can pass through screens as follows:

Screen, U. S. Standard Sieve Series	Percent
100-mesh	100
140-mesh	98 ± 2
200-mesh	90 ± 2
325-mesh	75 ± 2

(c) Chemical analysis of the sand and dust shall be as follows:

Substance	Percent by weight
SiO_2	97 to 99
Fe_2O_3	0 to 2
Al_2O_3	0 to 1
TiO_2	0 to 2
MgO	0 to 1
Ignition losses	0 to 2

(d) Maintain internal temperature of test chamber at 77° ± 4° F for a period of 6 hours, with sand and dust velocity through the test chamber between 100 and 500 feet per minute.

(e) After 6 hours at above conditions, raise the temperature to 160° F and maintain it at this value for 6 hours. The other conditions shall remain the same.

(f) Remove equipment from test chamber and allow it to cool to room temperature. Accumulated dust shall be removed by brushing, wiping or shaking, with care taken to avoid introduction of additional dust into the equipment. However, dust shall not be removed by either blast or vacuum cleaning. The equipment shall meet specified performance required in Para. 3.18.

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4.10.6 Noise level (see 3.2.4.2).- The assemblage shall be tested for compliance with 3.2.4.2 using the following instrumentation and procedures:

a. Instrumentation.-

(1) An octave-band noise analyzer containing an octave-band filter set meeting the American Standards Association (ASA) Specification Z24.10-1953 or an octave-band noise analyzer containing an octave-band filter set meeting the International Organization for Standardization's (ISO) recommendation ISO/R266-1962 shall be used.

(2) Instrumentation including ancillary items used to conduct tests shall be listed by manufacturer's name and model number in the test report.

b. Procedure.-

(1) All doors, windows, vents or other openings of the assemblage enclosure (shelter or van) shall be in a closed position unless required to be open for operation of the assemblage.

(2) The microphone of the test set shall be placed at the normal position of the operator's head. When more than one operator is normally used in the operation of the equipment, additional measurements shall be made at each position.

(3) Only those personnel required for normal operation of the equipment shall be within the enclosure during measurements.

(4) Octave-band pressure level measurements shall not excee that of the model.

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4.10.7 Test for internal vibration (see 3.28.7).- Internal vibration of Power Supply PP-4034( )/TRQ-23 shall be measured as follows, to determine conformance to requirements of 3.18. (This test shall apply to Power Supply PP-4034( )/TRQ-23.) Other components are to be tested under seperate specifications.

(a) Secure the equipment (shock mounts, if any, removed or blocked) directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be no less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.10.8 Test for equipment vibration (see 3.28.7.2).- Power Supply PP-4034( )/TRQ-23

(a) Secure the equipment on its shock mounts to a vibration table that can be controlled within 10 percent of the specified amplitude. The equipment shall be mounted in its normal operating position with connections (such as cable assemblies) in place.

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(b) Vibrate the equipment successively, in three mutually perpendicular directions, over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be 0.030 inch; however, this may be reduced, if necessary, to prevent excessive excursion at or near resonance provided that the excursion can still be measured.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure amplitude of the applied vibration and the vibration of the equipment, for frequencies between 10 and 25 and between 35 and 55 cycles per second.

(e) Determine the natural frequency of the equipment; the frequency at which the ratio of the equipment vibration to the applied vibration is greatest shall be considered the natural frequency.

4.10.9 Bounce test; class L units (see 3.28.7.3).- The Power Supply PP-4034( )/TRQ-23 shall be tested on the PackageTester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439 and shall be tested as follows:

(a) Secure the equipment to the 48-by 48-inch vehicular adapter plate, in the same manner as it would be mounted in a vehicle. The adapter plate shall be free of cracks and broken welds, and all unused holes except the 2-inch breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate.

(b) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

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(c) Place the adapter plate on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the backstop is not restrained.

(d) Operate the package tester, shafts in phase, for a total of 3 hours at 238 ± 1 rpm. Turn the adapter plate 90 degrees at the end of each 45 minutes, each time in the same direction. Rest periods may be provided to permit cooling of the resilient mounts, provided that the testing time is not less than 3 hours.

4.10.10 Bounce test; class H units (see 3.28.7.3).- A complete AN/TRQ-23 with equipment stored for transit, mounted on a Truck M-37 3/4 ton shall be driven 5 times over sections of the Munson test course at the Aberdeen Proving Grounds, Aberdeen, Maryland, or approved equal, in the order and at the speeds shown in Table VI.

TABLE VI - BOUNCE TEST

Section of Munson Test Course	Miles/Hour
Coarse washboard (6-inch waves spaced 72 inches apart)	5
Belgian block	20
Radial washboard (2-inch to 4-inch waves)	15
Single corrugation (4-inch to 6-inch waves)	20
Any short sections between the above sections	20

The set shall show no physical damage and shall be tested for and meet full requirements of 4.7.

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4.10.11 Shock test; railroad humping (see 3.28.7.3).- The test shall be as follows:

(a) Load the shelter--including contents--on a flat car, in the manner normally used for shipment. The contents shall include the items covered by this specification, plus any other items specified by the contract for use in this test. No bracing, dunnage, blocking, or other protective means shall be used inside the shelter, unless a permanent part of the equipment. The shelter shall be secured to the flat car, as specified in Specification MIL-S-52060A.

(b) Couple the flat car to two other cars. All three cars shall be stationary, with brakes off.

(c) Impact a 165,000 pound car, traveling at least 9 miles per hour, against the test car. For shelters which can only be positioned longitudinally, two impacts shall be performed on each end of the car with the shelter positioned laterally - four in all. The order in which these impacts are performed shall be selected by the contracting officer.

(d) Repeat the impact if the blocking which holds the shelter in place is torn loose.

(e) The equipment shall show no physical damage and shall meet requirements of 4.7.

4.10.11.1 Shock test; bench-handling (see 3.28.7.3).- The chassis-and front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal faces as axes, for a total of four drops.

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(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.10.12 Shelter weight distribution (see 3.7).-

(a) Fore and aft.- Locate shelter on level surface. Place a scale under one end and a block under the other end in a line with the center of gravity so shelter balances and is level in both lateral and longitudinal planes. Measure both fore and aft weights using the above procedure.

(b) Curb and Roadsides.- Place a scale under one side and a block under the other side in a line with the center of gravity so the shelter balances and is level in both lateral and longitudinal planes. Measure both curb and roadside weights using the above procedure.

(c) The shelter shall not exceed requirements in 3.7.

4.10.13 Weight test spare part case (see 3.21).- Prior to mounting the spare parts case in the shelter, two 250 pound weights shall be placed simulaneously on each of the case lids. The case shall exhibit no physical damage as a result of this test.

4.10.14 Road test (see 3.28.7.3).- The AN/TRQ-23 with all equipments stored for transit shall be driven over the Perryman Course, Aberdeen Proving Grounds, Aberdeen, Maryland, or an approved equal, for a total of 100 miles. Twenty laps of 5 miles each at 25 to 30 miles per hour; 10 laps counterclockwise and 10 laps clockwise interspersed. After testing the AN/TRQ-23 shall exhibit no physical damage and shall meet performance requirements of 4.7.

4.10.15 Inspection for dimensional interchangeability (see 3.28.8).- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.28.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

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(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimension of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assemblies or major units.

(e) Size and form of special threads.

4.10.16 Orientation test (see 3.28.4).- The AN/TRQ-23 shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left, and right) to an angel of 20^{+3}_{-1} degrees. During inclination in each plane, the systems shall function satisfactorily.

4.10.17 Electromagnetic compatibility (see 3.28.9).- The AN/TRQ-23 system shall be subjected to the applicable electromagnetic compatibility test of Specification MIL-E-55301.

4.10.18 Final system operational test (see 3.25).- Prior to preparation for delivery, each AN/TRQ-23 set shall be subjected to a 56 hour operational test. This test shall consist of the following:

(a) On each of three days during normal working hours, each component of the AN/TRQ-23 shall be operated under normal room ambient conditions and nominal input voltage to check all modes of operation. All active electronic components shall be turned on at the beginning of the test and every combination of antenna, antenna drive, receiver, patching, etc., shall be checked for operation on externally generated radiating signals.

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(b) All units except the heater, gyro and recorder reproducer shall be continuously powered during this 56 hour period and (except during gyro check period) shall not be turned off at any time. Components which do not operate at the end of the 56 hour period or which have not operated for the full 56 hours will not be accepted.

4.11 Visual and mechanical inspection.- The units shall be examined for the defects listed in Standard MIL-STD-252 and Table VII.

TABLE VII - CLASSIFICATION OF VISUAL AND MECHANICAL DEFECTS

Classification	Defect
MAJOR	Stay chain missing or broken. Carrying handle, hinge, or lid fastener attached to wrong side of case. Cover missing or inoperative. Control inoperative. Loose fit of antenna elements. Loose fit of compass to tripod. Inoperative or loose clamping action on adjustable antenna elements. Shelter mounting brackets missing, loose, bent or improperly installed. Unit tie-down facilities in shelter missing or not securely fastened.
MINOR	Cover reinforcement missing or improperly assembled. Difficulty in getting good contact in a control, or in making an accurate setting due to backlash or play. Control does not operate smoothly and freely. Loose fit of mast section

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4.12 Acceptance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine comformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Receiving Set, Radio AN/TRQ-23( ) shall be preserved, packaged, and inspected in accordance with procedures specified for the designated methods prescribed in MIL-P-116, as described in the following paragraphs.

5.1.1.1 Technical manual.- Each technical manual shall be packaged method C-1.

5.1.1.2 Spare electron tubes.- Each spare tube shall be packaged in accordance with the applicable provisions of MIL-E-75.

5.1.1.3 Spare fuses and lamps.- Package like stock numbered fuses five per unit package and package lamps individually in accordance with method III as follows: Cushion the items by wrapping in paperboard conforming to PPP-P-291, type I, style 2. Secure the wraps with gummed paper tape conforming to UU-T-111.

5.1.1.4 Miscellaneous spare parts.- Miscellaneous spare parts shall be preserved and packaged in accordance with the applicable provisions of MIL-E-17555.

5.1.1.5 Wire and cable assigned reels.- Wind the wire and cables on the assigned reels. Secure the wire and cable on the reels.

5.1.1.6 Wire, cords and cable not on assigned reels.- Wind each wire, cord and cable into a coil of the smallest proportionate dimensions and tie in three places with cotton tape conforming to DDD-T-86.

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5.1.1.7 Miscellaneous equipment and accessories.- Each miscellaneous equipment and accessory shall be packaged individually in accordance with method III, following the procedures specified in paragraph 5.1.1.3.

5.1.1.8 Clock.- Package each clock in accordance with method 1A-8 as follows: Cushion each clock by wrapping in paperboard conforming to PPP-P-291, type 1, style 2. Secure the wrap with gummed paper tape conforming to UU-T-111. Place the cushioned clock within a barrier bag conforming to MIL-B-117, class e. Place the water-vaporproofed clock within a close-fitting box conforming to PPP-B-566, style II, type A, class a. Box closure shall be made in accordance with the appendix of the box specification.

5.1.1.9 Cabinet and drawer stowage within shelter.- Stow the specified items within the designated compartments of the cabinets or drawers to the maximum capacity.

5.1.1.10 Overflow boxes.- Packaged items, remaining after filling the cabinets and drawers, shall be packed in nailed wood or corrugated triple-wall fiberboard boxes conforming to PPP-B-621, style 4, class 1 or PPP-B-640, class 1, grade A, style E, respectively. Like items shall be intermediately packaged in paperboard or fiberboard boxes prior to stowage within overflow boxes. Block and brace contents of overflow boxes in a manner that will prevent movement during transit. Box closure shall be made in accordance with the appendix of the applicable box specification.

5.1.1.11 Floor stowage.- All equipment and accessories not specifically assigned a location in the shelter shall be stowed and secured on the shelter floor.

5.1.1.12 Major components within shelter.- Major components assigned a permanent position within the shelter shall be carefully mounted and secured in position in accordance with the latest assembly drawings. Secure all fastenings, shock mounts, bolts, nuts, screws and straps of all major and auxilliary equipment mounted on the floor, shelves or walls of the shelter. Place and secure all covers.

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5.1.1.13 Interior shelter blocking.- Shock mounted equipment shall not be blocked or braced but shall float free. Design and construct a wood blocking and bracing assembly to support the stowed load and prevent shifting in transit. All surfaces of the wood blocking contacting the equipment shall be cushioned with paperboard conforming to PPP-P-291, type III, style 2.

5.1.1.14 Shelter with contents.- As applicable, each shelter with contents shall be packaged method III as follows:

5.1.1.14.1 Cable connectors.- Position caps over cable connectors and secure.

5.1.1.14.2 Shelter windows.- Close and secure all windows.

5.1.1.14.3 Blackout blinds.- Close and secure all blackout blinds.

5.1.1.14.4 Louver vents.- Close and secure the shields over the louver vents.

5.1.1.14.5 Blower exhaust vents.- Close and secure the shields over the blower vents.

5.1.1.14.6 Heating intake.- Close and secure the heating intake door.

5.1.1.14.7 Fire extinguisher.- Position and secure fire extinguisher.

5.1.1.14.8 Shelter door.- Close and secure the shelter doors. Further secure the doors with padlocks provided.

5.1.1.14.9 Padlock keys.- Place the padlock keys within a cloth bag conforming to PPP-B-20. Close the bag with the drawstring and secure by knotting. Secure the drawstring bag to the door hasp with number 15 AWG galvanized wire. Further secure the bag to the shelter by overlapping the bag with pressure-sensitive tape conforming to PPP-T-76.

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5.1.1.14.10 Mail slot.- Close and secure mail slot covers.

5.1.2 Level C.- Receiving Set, Radio AN/TRQ-23 shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage from the supply source to the first receiving activity.

5.1.2.1 Spare parts.- Spare parts items shall be preserved, packaged and stowed in accordance with the requirements cited in 5.1.1.2, 5.1.1.3, 5.1.1.4, 5.1.1.9 and 5.1.1.10.

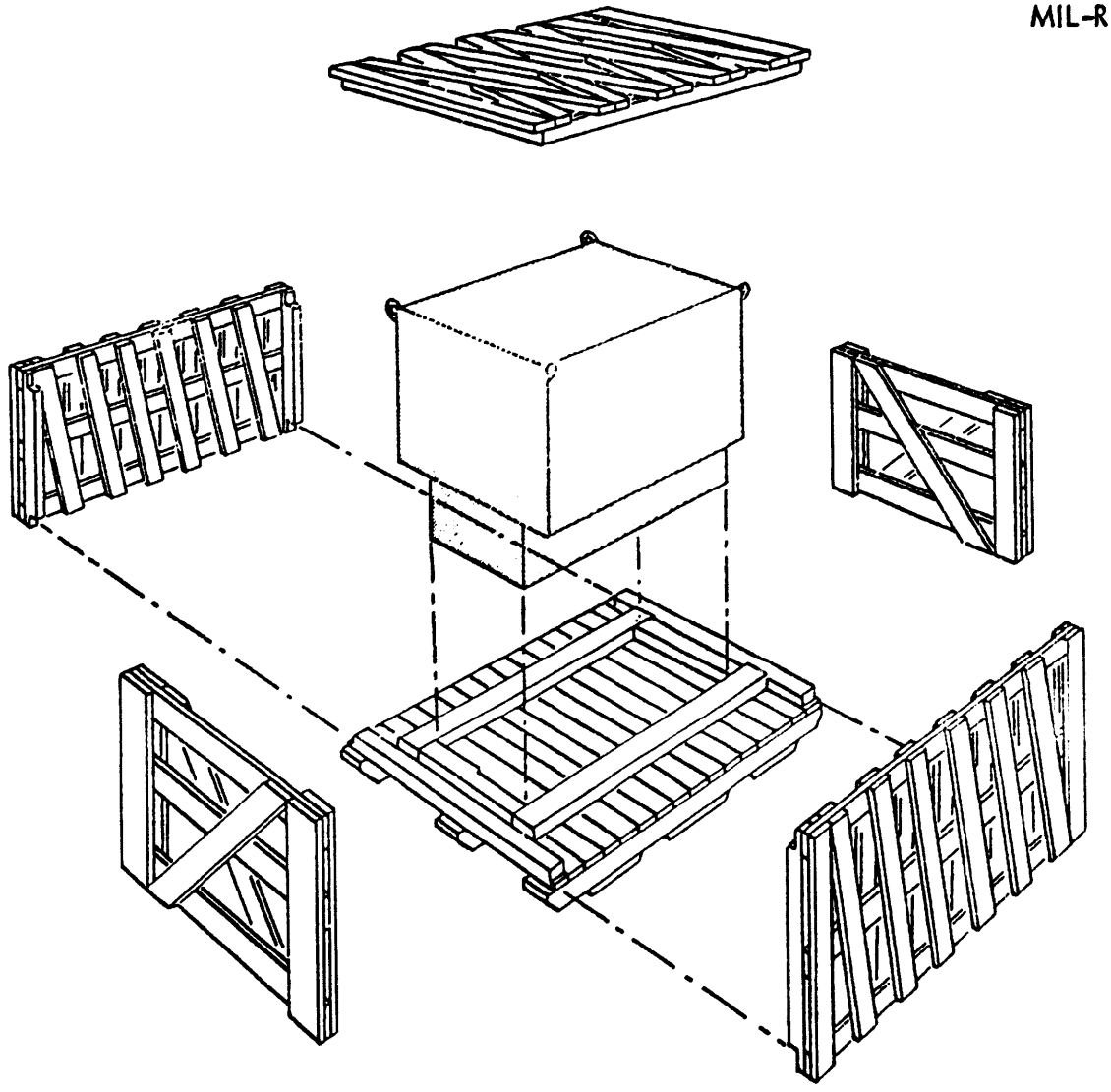
5.2 Packing.-

5.2.1 Level A.- Each shelter, containing Receiving Set, Radio AN/TRQ-23, shall be packed in a crate conforming to MIL-C-3774 (bolted, demountable). The skid base shall be modified to include blocking (4" x 4" wood members); illustrated in Figure 4, and secured to the floor members by nailing. Place and secure U bolts in position as illustrated in Figure 5. The U bolts shall be of sufficient length to extend through the skids of the mounting base and to accommodate washers and nuts. The U bolt threads shall be upset and coated with white lead paint to prevent turning and corrosion. All nuts and bolts on the under side of skids shall be counter-sunk flush to prevent friction or drag. Rubbing strips shall be applied to the crate skids in a manner that will accomodate fork lift entry. The shelter shall be secured to the skid base in the following manner: Position the shelter skid base. Blocking and bracing of shelter to skid base shall be as illustrated in Figure 4. Attach the hook of the sling assembly through the lifting device eye at the shelter top corners and secure other end of the sling assembly to the U bolts in the skid base (see Figure 5) maintaining a 45° angle of the slings. The slings shall be made taut by the use of turn buckles. Assemble the remaining crate components over the shelter.

5.2.1.1 Blocking, bracing and securing of crated shelters shall be as illustrated in Figure 6.

5.2.2 Level C.- Each shelter, containing Receiving Set, Radio AN/TRQ-23, shall be packed as specified in 5.2.1.

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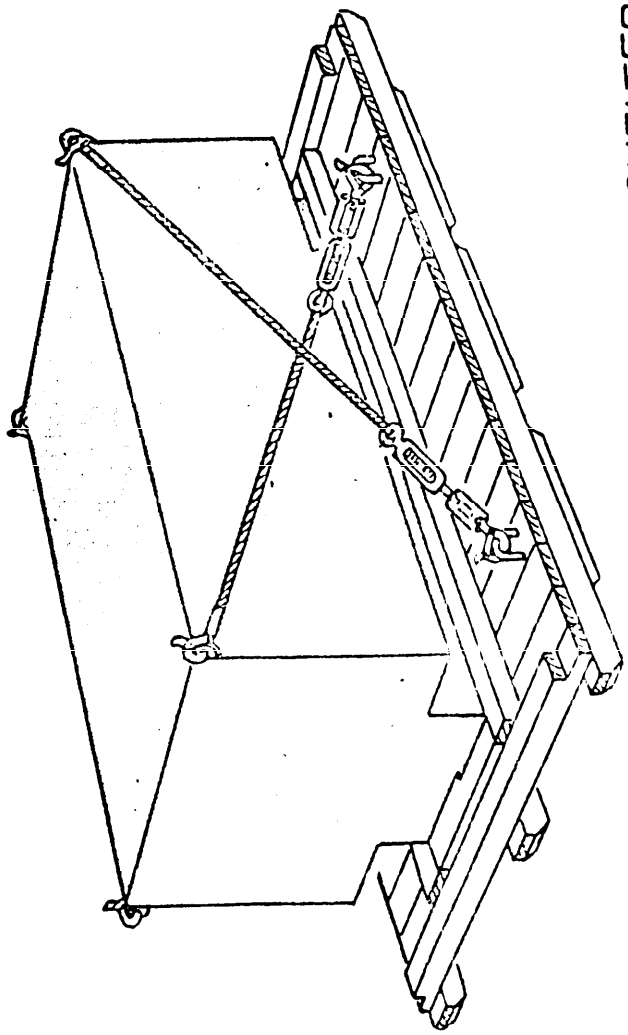


SHELTER CRATE

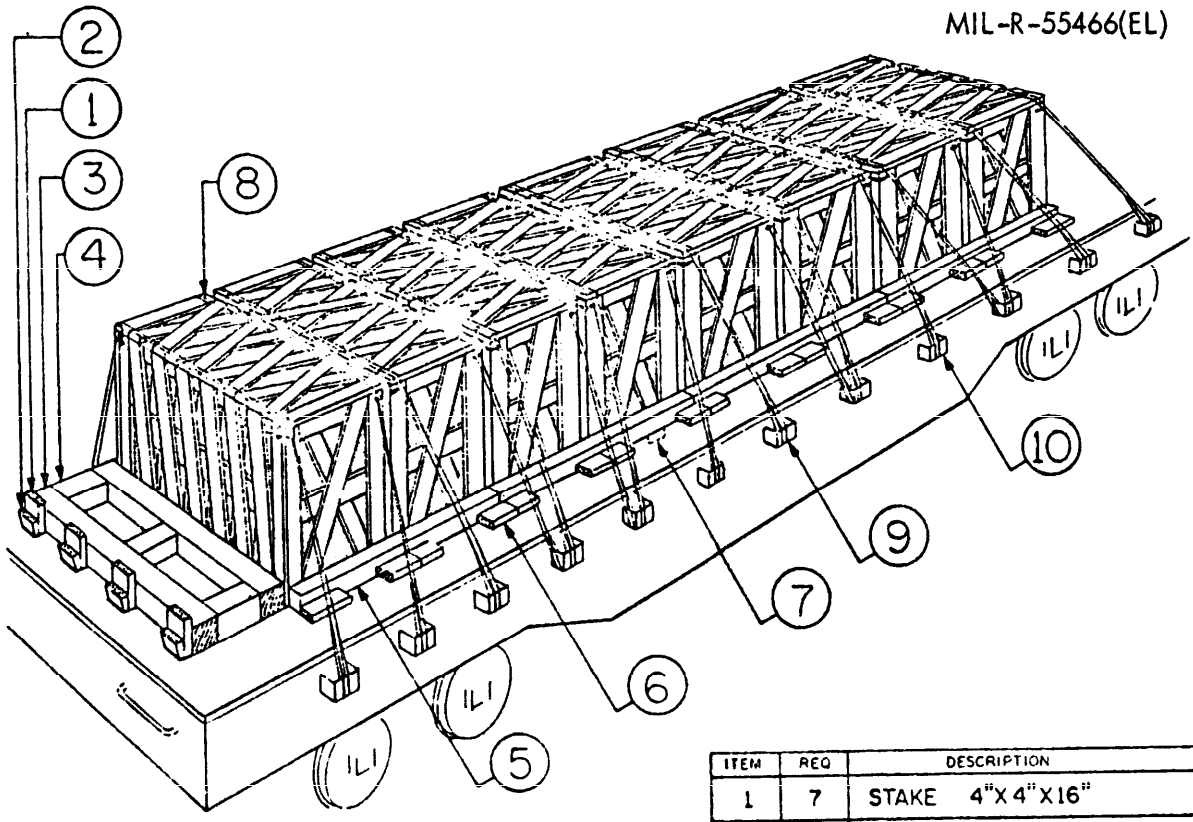
FIGURE 4

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FIGURE 5 TIE DOWN SHELTER



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ITEM	REQ	DESCRIPTION
1	7	STAKE 4"X4"X16"
2	7	STAKE WEDGE TO SUIT
3	4	LUMBER 6"X6" TO SUIT
4	5	LUMBER 6"X6" TO SUIT
5	8	LUMBER 4"X4"X12' OR TO SUIT
6	32	LUMBER 2"X4"X12"
7	18	PLATE ANTI SKID 4"X4" ATTACHED TO SKID BOTTOM
8	48	STAPLES 1½" L X1" GAP
9	64	8 SINGLE STRAND 2000 TEST 2 WIRES EA CORNER OF CRATE TO STAKE POCKETS
10	24	FLEX METAL SHEETING

SHELTER CRATED

NOTE:
(a) FLEXIBLE METAL SHEETING (0.042 INCH THICK) CUT TO SIZE SHALL BE PLACED IN STAKE POCKETS, I.E., SECURING POINT OR SECURING DEVICE FOR METAL WIRE TIE DOWNS AT POINTS OF CONTACT. UPON TIGHTENING METAL WIRE, SHEETING WILL CONTOUR TO A PROTECTIVE HOUSING FOR THE METAL WIRE TIE DOWN.
(b) CRATES ON FLATCAR CONTAINING SHELTER S-144 SHALL BE BUTTED AT SIDES.

FIGURE 6

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5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of MIL-STD-129 and the general requirements of paragraph 60 of the Appendix to MIL-C-3774 for crates. In addition, MIL-C-3774 crates will have the following notice stenciled on exterior sides and ends with letters three inches high: "BOLTED DEMOUNTABLE - REUSABLE CRATE - DO NOT DESTROY - USER TO RETAIN TIE DOWN SLINGS".

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, Level B, or Level C)

(d) The specific paragraphs of Section 5 which are applicable to the particular procurement.

(e) Production inspection:

(1) Three sample units of each item cited in Section 1 are generally required so that lengthy environmental tests can be completed on two sample units while complete performance measurements can be made on the third sample unit. (See 3.2)

(2) Preproduction Pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

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(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required. (See 3.28.11)

(i) Quantity of tools and running spare parts required. (See 3.28.11)

6.2 Color.- The color chips furnished by the procuring agency shall be in accordance with 3.5.2 and may be obtained upon request to Commanding Officer, U. S. Naval Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

6.3 Definition of Level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.5 Location of operational inspection.- It is desirable that the operational inspection (4.10.18) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: R-1418W/GR. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

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6.7 Color of textiles.- Standard samples of Olive Drab No. 7 may be obtained from Sample Loan Unit, Military Clothing and Textile Supply Agency, Philadelphia, Pennsylvania. When requesting samples, indication should be given as to type of material concerned.

6.8 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provision established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN
Army (EL)

PREPARING ACTIVITY
Army (EL)

PROJECT NUMBER
5895-A079

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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HEADQUARTERS
U. S. ARMY ELECTRONICS COMMAND
FORT MONMOUTH, NEW JERSEY 07703

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OFFICIAL BUSINESS

Commanding General
U. S. Army Electronics Command
ATTN: AMSEL-PP-

Fort Monmouth, New Jersey 07703

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